

MIS: Diagnosing Structural Bottlenecks in Copy-Paste Augmentation for Agricultural Object Detection

Huy Khanh Tran, Yuichiro Nomura, and Hiroshi Mineno
Graduate School of Integrated Science and Technology, Shizuoka University, Japan
tran.huy.khanh.25@shizuoka.ac.jp

Abstract—Copy-paste augmentation achieves strong results for instance segmentation yet yields inconsistent or negative gains for bounding-box detection in agricultural scenes. We propose the Minimum Inference Score (MIS), a difficulty metric derived from a trained baseline detector that correlates strongly with detection failure modes across objects and images. MIS reveals where and how a model fails on a given dataset: when hard instances concentrate in structurally constrained regimes—driven by object size, domain composition, or both—copy-paste augmentation cannot provide reliable gains. MIS also predicts selection strategy risk: outcome variance scales with sourcedomain difficulty, a pattern readable from partition composition before any augmented run. We validate both findings across three agricultural benchmarks, where MIS partition analysis identifies structural constraints that predict copy-paste failure, and positive gains in their absence.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural object detection, data augmentation diagnostic

I. INTRODUCTION

Copy-paste augmentation composites cropped labeled instances into training images to increase sampling diversity [1], [2]. While effective for instance segmentation [2], bounding-box detection in agricultural imagery is subject to harsher constraints—sub-resolution instances, multi-domain acquisition, and high object density—frequently producing inconsistent or negative augmentation outcomes and forcing costly multi-seed search without a principled stopping criterion.

We introduce the **Minimum Inference Score (MIS)**, a metric computed from a trained baseline detector that captures model failure patterns on a given dataset. By correlating with miss rate and FP rate, MIS surfaces where difficulty concentrates in a dataset (driven by object size, domain composition, or their interaction), and enables two practical outputs: determining whether copy-paste can provide useful training signal, and predicting relative selection strategy risk from partition composition alone. On MinneApple [3], GWHD 2021 [4], and WGISD [5], MIS partition analysis identifies structural constraints that predict copy-paste failure, and positive gains when no such constraint is present, providing bidirectional diagnostic validation.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ denote ground-truth boxes in image I . A trained detector produces predictions p with confidence c_p at $\tau_{\text{conf}}=0.001$ to retain near-miss candidates. For each $g \in G(I)$, let $\mathcal{P}(g)=\{p : \text{IoU}(p, g) \geq 0.5\}$. The *Object Difficulty Score* is

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset \\ \alpha + \beta, & \mathcal{P}(g) = \emptyset \end{cases} \quad (1)$$

with $\alpha=\beta=0.5$ and $u_p=\text{IoU}(p, g)$. Unmatched objects receive the maximum score ($\alpha+\beta=1$). The *Image Difficulty Score* is

$$S_{\text{img}}(I) = \frac{1}{|G(I)|} \sum_g S_{\text{obj}}(g) + w_2 \cdot \text{FPRate}(I), \quad (2)$$

where $\text{FPRate}(I)$ counts unmatched predictions per image. This yields $S_{\text{obj}} \in [0, 1]$ for every ground-truth object and S_{img} for every training image, where higher values indicate greater detection difficulty. By construction, S_{img} encodes both axes of detection failure (miss rate and FP rate); w_2 is set by grid search to balance their contributions. MIS scores are model-conditioned and reflect structural properties through detector failure behavior, not directly.

B. Augmentation Protocol

We train a YOLOv8s baseline and partition all training objects by S_{obj} and all training images by S_{img} into equal difficulty tertiles (Low, Medium, High). Four copy-paste conditions are evaluated: **Score High** pastes objects from the highest-difficulty object tertile into the highest-difficulty destination images; **Score Medium** and **Score Low** follow the same logic for their respective tertiles. **Random** selects both destination images and source objects without score-based filtering. This design isolates whether directing augmentation toward specific difficulty regimes affects outcome. Following [1], training data is expanded by 30% via unblended within-image copy-paste: 2–8 objects per image, scale $[0.9, 1.1]$, rotation $\pm 15^\circ$, $3\times$ maximum reuse; all other settings follow the default YOLOv8 protocol.

TABLE I
MINNEAPPLE: S_{obj} CAPTURES SIZE-LEVEL DIFFICULTY.

Category	AP	Mean S_{obj}	Top tertile
Very Small ($\leq 400 \text{ px}^2$)	0.071	0.152	64.4%
Small (400–1024 px^2)	0.270	0.076	33.9%
Medium ($> 1024 \text{ px}^2$)	0.493	0.053	1.7%

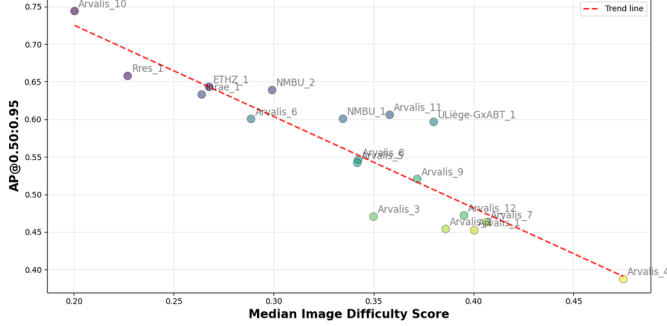


Fig. 1. GWHD 2021: per-domain AP vs. median image difficulty (MIS).

III. RESULTS

A. MIS Reflects Detection Failure Modes

Pearson correlations between S_{img} and miss rate on the validation set are $r=0.79, 0.81, 0.91$ on MinneApple, GWHD, and WGISD; correlations with FP rate are $r=0.79, 0.81, 0.84$. Strong correlations across all three datasets confirm that MIS captures the primary axes of detection failure as intended by its construction.

B. MIS Identifies Structural Failure Sources

Size (MinneApple). S_{obj} inversely tracks object area across all training instances (Spearman $r_s=-0.82$; Table I): 64.4% of the top-difficulty tertile falls in *Very Small* instances ($\leq 400 \text{ px}^2$), where AP is near-zero (0.071), indicating hard objects concentrate below the model’s size recognition threshold.

Domain (GWHD 2021). Per-domain median MIS strongly anticorrelates with domain AP (Pearson $r=-0.90$; Fig. 1): MIS correctly ranks domain difficulty from detector behavior alone, regardless of its underlying cause—varied cameras, growth stages, and geographic acquisition conditions.

C. Application: Pre-Augmentation Diagnostic for Copy-Paste

MinneApple (resolution ceiling). MIS identifies hard objects concentrated below the recognition threshold. All four copy-paste conditions yield negative ΔAP (Table II), confirming that additional sub-resolution instances provide no usable learning signal regardless of selection strategy.

GWHD 2021 (domain difficulty). MIS partitioning reveals near-complete domain separation: low-difficulty images are 66% in ETHZ_1/Rres_1 domain with ($\text{AP} \approx 0.65$), while high-difficulty images are 78% Arvalis variants ($\text{AP} \approx 0.46$). This structure directly predicts per-condition behavior. Score Low sources from high-AP domains, reinforcing already well-learned patterns which result in consistent, low-variance drop

TABLE II
COPY-PASTE OUTCOMES (3-SEED MEAN $\pm \sigma$). BOTTLENECK: STRUCTURAL FAILURE SOURCE IDENTIFIED BY MIS.

Dataset	Bottleneck	Base AP	Best Δ	Worst Δ
MinneApple	Size	0.353 ± 0.005	-0.003 (Hi)	-0.008 (Rnd)
GWHD 2021	Domain	0.229 ± 0.013	-0.006^* (Hi)	-0.034 (Lo)
WGISD	—	0.480 ± 0.007	$+0.009$ (Med)	$+0.001$ (Hi)

*Score High on GWHD: $\sigma=0.040$, high seed variance.

($-0.034, \sigma=0.005$). Score High sources from hard Arvalis domains where difficulty is acquisition-based; added instances cannot resolve domain gaps, yielding high instability ($-0.006, \sigma=0.040$). Random and Medium sample without domain awareness, producing moderate drops ($-0.015, -0.017$) at intermediate variance ($\sigma=0.023, 0.021$). Both outcome direction and variance scale with sourced-domain difficulty—patterns MIS predicts from partition composition alone, confirming a structural bottleneck no selection strategy can overcome.

WGISD (positive control). MIS raises no flag; all conditions confirm viability with positive gains (Table II), with Score Medium achieving the best result.

Across all three benchmarks, the *direction* of copy-paste outcome is consistent with the MIS diagnostic: all conditions are negative on flagged datasets and positive on the unflagged dataset. This directional consistency, independent of selection strategy, supports a structural rather than selection-driven explanation and directly addresses why copy-paste struggles: it cannot overcome the underlying difficulty MIS identifies.

IV. CONCLUSION

MIS links baseline detector failure behavior to difficulty concentration patterns, enabling a practical pre-augmentation signal that indicates whether copy-paste is likely to yield reliable gains. Furthermore, outcome variance across GWHD conditions scales monotonically with sourced-domain difficulty, suggesting MIS partitioning can inform selection strategy risk even when augmentation is inadvisable. In our experiments, 24 augmented training runs across MIS-flagged datasets (MinneApple and GWHD, 4 conditions $\times$ 3 seeds each) yielded zero reliable gains; MIS identifies these datasets without any augmented run, directly saving that compute budget. Practitioners should apply MIS after baseline training: when MIS identifies strong or moderate concentration of high-difficulty instances in near-zero AP regimes, copy-paste augmentation is unlikely to yield reliable gains. This finding is validated bidirectionally across three agricultural benchmarks. Future work includes using MIS to guide instance-level selection when structural constraints are absent, and combining MIS with generative methods to produce information-rich training instances for hard regimes.

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